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Xin Di (邸新), PhD

Research Assistant Professor, Department of Biomedical Engineering

New Jersey Institute of Technology, Newark, NJ, USA

✉ xin.di@njit.edu | 🌐 www.dixin.info | Google Scholar: wDjD46gAAAAJ

ACEDEMIC APPOINTMENT

- Research Assistant Professor, Department of Biomedical Engineering, New Jersey Institute of Technology, Newark, NJ, USA (2012 – present)
- Associate Professor (Protocol Appointment), Department of Biomedical Engineering, University of Electronic Science and Technology of China, Chengdu, China (2019 – 2020)
- Postdoctoral Researcher, Department of Radiology, University of Medicine and Dentistry of New Jersey, Newark, NJ, USA (2011 – 2012)

EDUCATION

- Ph.D. in Psychology, Sun Yat-Sen University, Guangzhou, China, 2010
- M.Sc. in Psychology, Sun Yat-Sen University, Guangzhou, China, 2006
- B.Eng. in Electronic Engineering, Civil Aviation University of China, Tianjin, China, 2004

RESEARCH FUNDING

- NJ Alliance for Clinical and Translational Science (NJ ACTS), PI: Xin Di | 2026 - 2027
  Establishing a Standardized Data Access Protocol (SDAP) to Resolve Data Fragmentation in the US Molecular Imaging Ecosystem
  Total cost: $40,000
- NJDOH CAUT25BRP005, PI: Xin Di | 2024 - 2026
  Exploring eye-movement patterns and brain activity in autism spectrum disorder through deep learning-based eye-tracking with functional MRI
  Total cost: $200,000

Completed Projects

- NIH R15MH125332, PI: Xin Di | 2021 - 2025
  Functional brain developments during movie watching and resting-state in autism spectrum disorder
  Total cost: $300,000
- NJ Alliance for Clinical and Translational Science (NJ ACTS), Co-PI: Xin Di | 2019 - 2020
  Towards development of stable multimodal neuroimaging based markers of AD progression
  Total cost: $20,000
- NJDOH CAUT16APL019, PI: Xin Di | 2016 - 2018
  Multimodal neuroimaging study of sex differences in children with autism spectrum disorder
  Total cost: $400,000

AWARDS

- Publons Top Reviewers for Neuroscience & Behavior (Top 1%), 2018
- Publons Top Reviewers for Neuroscience (Top 1%), 2017
- Publons Sentinels of Science Awards (2nd in neuroscience), 2016
- Travel stipend, ISMRM Annual Meeting, Toronto (2008); Berlin (2007)

PROFESSIONAL ACTIVITIES

Symposium and Conference Organization

- Organizer, *Relationship between Brain Energy Consumption and Brain Organizations*, OHBM, Brisbane, 2025

Scientific Committees

- Chair, Outreach Council, *Molecular Connectivity Working Group*, 2022 – present
  - Organizer, *Molecular Connectivity Online Series (MCOS)*
- Organizing Committee, *Neurochat Online Conference*, 2020 – 2021

Editorial Boards

- Associate Editor, *Frontiers Brain Imaging Methods*, 2020 – present
- Editorial Board Member, *BMC Neuroscience*, 2020 – present
- Review Editor, various Frontiers sections, 2014 – present

Reviewer

- *Ad hoc journal reviewer* for 40+ journals, including Nature Communications, Cerebral Cortex, and Imaging Neuroscience (full list available on request)
- *Grant reviewer*, NIH CPDD Study Section (June & Oct 2024); Ontario Mental Health Foundation (2013)
- *Conference abstract reviewer*, OHBM (2013–2015); Resting State / Brain Connectivity (2018)

PUBLICATIONS (Google Scholar h-index: 37)

Working Papers

6. **Di X**, Biswal BB (2026). Context-Specific Decoupling and Competing Phenotypes: Transdiagnostic Eye-Tracking Biomarkers of ASD and ADHD During Naturalistic Viewing in a Large Pediatric Cohort. bioRxiv: 10.64898/2026.05.11.724367
5. **Di X**, Xu T, Castellanos FX, Biswal BB (2026). Unique Amygdala Signatures and Shared Prefrontal Deficits in Autism: Mapping Social Heterogeneity via Naturalistic functional Magnetic Resonance Imaging. bioRxiv: 10.64898/2026.02.26.708280
4. Gao L, Wei Z, Biswal BB, **Di X** (2025). fMRI-Based Prediction of Eye Gaze During Naturalistic Movie Viewing Reveals Eye-Movement-Related Brain Activity. bioRxiv: 10.64898/2026.01.10.698820
3. **Di X**, Hanna GB, Biswal BB (2025). Timing Is Everything: Temporal Dynamics of BOLD Responses to Naturalistic Features in Movie Watching. bioRxiv: 10.1101/2025.11.07.687226

2. Chen DY, **Di X**, Karunakaran KD, Sun H, Pal S, Biswal BB (2024). Delayed cerebrovascular reactivity in individuals with spinal cord injury in the right inferior parietal lobe: a breath-hold functional near-infrared spectroscopy study. medRxiv: 10.1101/2024.06.03.24307819
1. Chen DY, **Di X**, Amaya N, Sun H, Pal S, Biswal BB (2024). Brain activation during the N-back working memory task in individuals with spinal cord injury: a functional near-infrared spectroscopy study. bioRxiv:10.1101/2024.02.09.579655

Reviews and Commentaries

6. Cavaliere C, et al. (2026). Molecular connectivity studies of cerebral glucose metabolism and blood flow: a scoping review. Neurosci Biobehav Rev 106623.
5. **Di X**, Xu T, Miao X (2026). Future perspectives of functional connectivity: Advances in data acquisition and analysis. In *Functional Connectivity of the Human Brain* (pp. 281-309). Academic Press.
4. Severino M, et al. (2025). Molecular connectivity studies in neurotransmission: a scoping review. Imaging Neuroscience, 3: imag_a_00530.
3. **Di X**, Zhang L, Zhang Z, Biswal BB (2024). Opportunities and Challenges in Connectivity Analysis for Task-Based fMRI. Phys Life Rev 51:13-17.
2. Hu Z, **Di X** and Yang Z (2023). Editorial: Shared responses and individual differences in the human brain during naturalistic stimulations. Front Hum Neurosci 17:1201728.
1. Qian C, **Di X** (2011). Phase or amplitude? The relationship between ongoing and evoked neural activity. J Neurosci 31(29):10425-10426.

Research Articles

68. **Di X**, Biswal BB, and Alzheimer's Disease Neuroimaging Initiative (accepted). Comparing Intra- and Inter-individual Correlational Brain Connectivity from Functional and Structural Neuroimaging Data. Brain Struct Funct 230:113.
67. **Di X**, Pratik Jain, Bharat B Biswal (accepted). Effects of Tasks on Functional Brain Connectivity Derived from Inter-Individual Correlations: Insights from Regional Homogeneity of Functional MRI Data. Brain Connect 15(6):207–216.
66. Galli A, Inglese M, Presotto L, Malito R, *Di X*, Toschi N, Pilotto A, Padovani A, Tassorelli C, Perani D, Sala A, Caminiti SP (2025). Glucose metabolism in hyper-connected regions predicts neurodegeneration and speed of conversion in Alzheimer's disease. Eur J Nucl Med Mol Imaging 52:4639–4651.
65. Jain P, Michael AM, Wang P, **Di X**, Biswal BB (accepted). WhiFuN: A Toolbox to Map the White Matter Functional Networks of the Human Brain. Imaging Neuroscience 3:IMAG.a.3.
64. Sohn W, **Di X**, Liang Z, Zhang Z, Biswal BB (2024). Explorations of using a convolutional neural network to understand brain activations during movie watching. Psychoradiology 4:kkae021.
63. Mittal P, Sao AK, Biswal BB, **Di X**, Dileep AD (2024). Network-wise analysis of movie-specific information in dynamic functional connectivity using COBE, Cereb Cortex 34(4):bhae170.
62. Chen DY, **Di X**, Yu X, Biswal BB (2024). The Significance and Limited Influence of Cerebrovascular Reactivity on Age and Sex Effects in Task- and Resting-State Brain Activity. Cereb Cortex 34(2): bhad448.

61. Chen DY, **Di X**, Biswal BB (2024). Neurovascular reactivity increases across development in the visual and frontal pole networks as revealed by a breath-holding task: a longitudinal fMRI study. Hum Brain Mapp 45(1):e26515.
60. Zhang H, Meng C, **Di X**, Wu X, Biswal BB (2023). Static and Dynamic Functional Connectome Reveals Reconfiguration Profiles of Whole Brain Network across Cognitive States. Netw Neurosci 7 (3):1034–1050.
59. **Di X**, Xu T, Uddin LQ Biswal BB (2023). Individual differences in time-varying and stationary brain connectivity during movie watching from childhood to early adulthood: Effects of age, sex, and behavioral associations. Dev Cogn Neurosci 63:101280.
58. Zhang H, **Di X**, Rypma B, Yang H, Meng C, Biswal BB (2023). Interaction between memory load and experimental design on brain connectivity and network topology. Neurosci Bull 39:631–644.
57. **Di X**, Biswal BB (2023). A functional MRI preprocessing and quality control protocol based on statistical parametric mapping (SPM) and MATLAB. Front Neuroimaging 1:1070151.
56. **Di X**, Zhang Z, Xu T, Biswal BB (2022): Dynamic and stationary brain connectivity during movie watching as revealed by functional MRI. Brain Struct Funct doi:10.1007/s00429-022-02522-w
55. Yang H, Zhang H, Meng C, Wohlschläger A, Brandl F, **Di X**, Tian L, Biswal BB (2022): Frequency-specific coactivation patterns in resting-state and their alterations in schizophrenia: an fMRI study. Hum Brain Mapping 43(12):3792-3808.
54. **Di X**, Woelfer M, Kuhn SB, Zhang Z, Biswal BB (2022). Estimations of the weather effects on brain functions using functional MRI - a cautionary tale. Hum Brain Mapping 43(11):3346-3356.
53. **Di X**, Biswal BB (2022). Principal component analysis reveals multiple consistent responses to naturalistic stimuli in children and adults. Hum Brain Mapp 43(11):3332-3345.
52. Li M, Danyeli LV, Colic L, Wagner G, Smesny S, Chand T, **Di X**, Biswal BB, Kaufmann J, Reichenbach JR, Speck O, Walter M, Sen ZD (2022). The differential association between local neurotransmitter levels and whole-brain resting-state functional connectivity in two distinct cingulate cortex subregions. Hum Brain Mapp 43(9):2833-2844.
51. Hafiz R, Gandhi TK, Mishra S, Prasad A, Mahajan V, **Di X**, Natelson BH, Biswal BB (2022). Higher Limbic and Basal Ganglia volumes in surviving COVID-negative patients and the relations to fatigue. Neuroimage:Reports 2(2):100095.
50. Li L, **Di X**, Zhang H, Huang G, Zhang L; Liang Z, Zhang Z (2022). Characterization of whole-brain task-modulated functional connectivity in response to nociceptive pain: A multisensory comparison study. Hum Brain Mapp 43(3):1061-1075.
49. Wu Z, Cao M, **Di X**, Wu K, Gao Y, Li X (2022). Regional Topological Aberrances of White Matter- and Gray Matter-based Functional Networks for Attention Processing may Foster Traumatic Brain Injury-Related Attention Deficits in Adults. Brain Sci 12(1), 16.
48. Yang H, Zhang H, **Di X**, Wang S, Meng C, Tian L, Biswal BB (2021). Reproducible Coactivation Patterns of Functional Brain Networks Reveal the Aberrant Dynamic State Transition in Schizophrenia. Neuroimage 237:118193.
47. **Di X**, Zhang Z, Biswal BB (2020). Understanding psychophysiological interaction and its relations to beta series correlation. Brain Imaging Behav, doi:10.1007/s11682-020-00304-8
46. Woelfer M, Li M, Colic L, Liebe T, **Di X**, Biswal B, Murrough J, Lessmann V, Brigadski T, Walter M (2020). Ketamine-induced changes in plasma brain-derived neurotrophic factor

(BDNF) levels are associated with the resting-state functional connectivity of the prefrontal cortex. World J Biol Psychia 21(9):696-710.
45. Klugah-Brown B, **Di X**, Zweerings J, Mathiak K, Becker B, Biswal B (2020): Common and separable neural alterations in substance use disorders: evidence from coordinate-based meta-analyses of functional neuroimaging studies in human. Hum Brain Mapp 41(16):4459-4477.
44. **Di X**, Biswal BB (2020). Intersubject consistent dynamic connectivity during natural vision revealed by functional MRI. Neuroimage 216:1166982.
43. Yang H, **Di X**, Gong Q, Sweeney J, Biswal BB (2020). Investigating inhibition deficit in schizophrenia using task-modulated brain networks. Brain Struct Funct 225:1601–1613.
42. Botvinik-Nezer et al., (2020): Variability in the analysis of a single neuroimaging dataset by many teams. Nature 582(7810):84-88.
41. **Di X**, Zhang H, Biswal BB (2020). Anterior cingulate cortex differently modulates fronto-parietal functional connectivity between resting-state and working memory tasks. Human Brain Mapping. Hum Brain Mapp 41:1797–1805.
40. **Di X**, Woelfer M, Amend M, Wehrl H, Ionescu TM, Pichler BJ, Biswal BB, and Alzheimer's Disease Neuroimaging Initiative (2019). Interregional causal influences of brain metabolic activity reveal the spread of aging effects during normal aging. Hum Brain Mapp 40(16):4657-4668.
39. Amend M, Ionescu TM, **Di X**, Pichler BJ, Biswal BB, Wehrl HF (2019). Functional resting-state brain connectivity is accompanied by dynamic correlations of application-dependent [18F]FDG PET-tracer fluctuations. Neuroimage 196:161-172.
38. Fu Z, Tu Y, **Di X**, Du Y, Sui J, Biswal BB, Zhang Z, de Lacy N, Calhoun V (2019). Transient Increased Thalamic-Sensory Connectivity and Decreased Whole-Brain Dynamism in Autism. Neuroimage 190:191-204.
37. **Di X**, Biswal BB (2019). Toward Task Connectomics: Examining Whole-Brain Task Modulated Connectivity in Different Task Domains. Cereb Cortex 29(4):1572-1583.
36. Fu Z, Tu Y, **Di X**, Du Y, Pearlson GD, Turner JA, Biswal BB, Zhang Z, Calhoun VD (2018). Characterizing Dynamic Amplitude of Low-Frequency Fluctuation and Its Relationship with Dynamic Functional Connectivity: An Application to Schizophrenia. Neuroimage 180:619-31.
35. **Di X**, Azeez A, Li X, Haque E, Biswal BB (2018). Disrupted focal white matter integrity in autism spectrum disorder: a voxel-based meta-analysis of diffusion tensor imaging studies. Prog Neuropsychopharmacol Biol Psychiatry 82:242-248.
34. Fu Z, Tu Y, **Di X**, Biswal BB, Calhoun VD, Zhang Z (2017). Associations between Functional Connectivity Dynamics and BOLD Dynamics are Heterogeneous across Brain Networks. Front Hum Neurosci 11:593.
33. **Di X**, Gohel S, Thielcke A, Wehrl HF, Biswal BB, and Alzheimer's Disease Neuroimaging Initiative (2017). Do all roads lead to Rome? A comparison of brain networks derived from inter-subject volumetric and metabolic covariance and moment-to-moment hemodynamic correlations in old individuals. Brain Struct Funct 222(8):3833–3845.
32. **Di X**, Biswal BB (2017). Psychophysiological Interactions in a Visual Checkerboard Task: Reproducibility, Reliability, and the Effects of Deconvolution. Front Neurosci 11:573.
31. Jin H, Wang P, Fang Z, **Di X**, Ye Z, Xu G, Lin H, Cheng Y, Li Y, Xu Y, Rao H (2017). Effects of badminton expertise on representational momentum: A combination of cross-sectional and longitudinal studies. Front Psychol 8:1526.

30. **Di X**, Reynolds RC, Biswal BB (2017). Imperfect (de)convolution may introduce spurious psychophysiological interactions and how to avoid it. Hum Brain Mapp 38(4), 1723–1740.
29. **Di X**, Huang J, Biswal BB (2017). Task-modulated brain connectivity of the amygdala: a meta-analysis of psychophysiological interactions. Brain Struct Funct 222(1):619-634.
28. Xu H, Wang P, Ye Z, **Di X**, Xu G, Mo L, Lin H, Rao H and Jin H (2016) The Role of Medial Frontal Cortex in Action Anticipation in Professional Badminton Players. Front Psychol. 7:1817.
27. Ray S, **Di X**, Biswal BB (2016). Effective Connectivity Within the Mesocorticolimbic System During Resting-State in Cocaine Users. Front Hum Neurosci. 10:563.
26. Yuan R, **Di X**, Taylor PA, Gohel S, Tsai YH, Biswal BB (2016). Functional topography of the thalamocortical system in human. Brain Struct Funct 221(4):1971-1984.
25. Zhang X, **Di X**, Lei H, Yang J, Xiao J, Wang X, Yao S, Rao H (2016): Imbalanced Spontaneous Brain Activity in Orbitofrontal-Insular Circuits in Individuals with Cognitive Vulnerability to Depression. J Affect Disord 198:56-63.
24. **Di X**, Biswal BB (2016). Similarly expanded bilateral temporal lobe volumes in female and male children with autism spectrum disorder. Biological Psychiatry: Cognitive Neuroscience and Neuroimaging 1(2):178-185.
23. Hu C, **Di X**, Eickhoff SB, Zhang M, Peng K, Guo H, Sui J (2016). Distinct and common aspects of physical and psychological self-representation in the brain: A meta-analysis of self-bias in facial and self-referential judgements. Neuroscience & Biobehavioral Reviews 61:197–207.
22. **Di X**, Biswal BB (2015). Characterizations of resting-state modulatory interactions in human brain. J Neurophysiol 114(5), 2785-96.
21. **Di X**, Fu Z, Chan SC, Hung YS, Biswal BB, Zhang Z (2015). Task-related Functional Connectivity Dynamics in a Block-designed Visual Experiment. Front Hum. Neurosci 9:543.
20. **Di X**, Biswal BB (2015). Dynamic Brain Functional Connectivity Modulated by Resting-State Networks. Brain Struct Funct 220(1):37-46.
19. **Di X**, Kim E, Chen P, Biswal BB (2014). Lateralized Resting-state Functional Connectivity in the Task-positive and Task-negative Networks. Brain Connect 4(9): 641-648.
18. Lei H, Zhang X, **Di X**, Rao H, Ming Q, Zhang J, Guo X, Jiang Y, Gao Y, Yi J, Zhu X, Yao S (2014). A Functional Polymorphism of the MAOA Gene Modulates Spontaneous Brain Activity in Pons. Biomed Res Int 2014:243280.
17. Fu Z, Chan SC, **Di X**, Biswal B, Zhang Z (2014). Adaptive Covariance Estimation of Non-stationary Processes and its Application to Infer Dynamic Connectivity from fMRI. IEEE Trans Biomed Circuits Syst 8(2):228–39.
16. **Di X**, Biswal BB (2014). Modulatory Interactions between the Default Mode Network and Task Positive Networks in Resting-State. PeerJ 2:e367.
15. **Di X**, Biswal BB (2014). Identifying the Default Mode Network Structure Using Dynamic Causal Modeling on Resting-state Functional Magnetic Resonance Imaging. Neuroimage 86:53–9.
14. **Di X**, Rypma B, Biswal BB (2014). Correspondence of Executive Function Related Functional and Anatomical Alterations in Aging Brain. Prog Neuropsychopharmacol Biol Psychiatry 48(3):41–50.

13. Yuan R, **Di X**, Kim EH, Barik S, Rypma B, Biswal BB (2013). Regional Homogeneity of Resting-state fMRI Contributes to Both Neurovascular and Task Activation Variations. Magn Reson Imaging 31(9):1492–1500.
12. **Di X**, Gohel S, Kim EH and Biswal BB (2013). Task vs. Rest - Different Network Configurations between the Coactivation and the Resting-State Brain Networks. Front Hum Neurosci. 7:493.
11. **Di X**, Biswal BB (2013). Modulatory interactions of resting-state brain functional connectivity. PLoS One 8(8): e71163.
10. **Di X**, Kim EH, Huang C, Tsai S, Lin C and Biswal BB (2013). The influence of the amplitude of low-frequency fluctuations on resting-state functional connectivity. Front Hum Neurosci. 7:118.
9. Huang J, Wang Y, Jin Z, **Di X**, Yang T, Gur RC, Gur RE, Shum DH, Cheung EF, Chan RC (2013). Happy facial expression processing with different social interaction cues: An fMRI study of individuals with schizotypal personality traits. Prog Neuropsychopharmacol Biol Psychiatry 44(1):108–17.
8. **Di X**, Kannurpatti SS, Rypma B, Biswal BB (2013). Calibrating BOLD fMRI activations with neuro-vascular and anatomical constraints. Cereb Cortex 23 (2):255-63.
7. **Di X**, Biswal BB, Alzheimer's Disease Neuroimaging Initiative (2012). Metabolic Brain Covariant Networks as Revealed by FDG-PET with reference to resting-state fMRI networks. Brain Connect 2(5):275-83.
6. **Di X**, Zhu S, Jin H, Wang P, Ye Z, Zhou K, Zhuo Y, Rao H (2012). Altered resting brain function and structure in professional badminton players. Brain Connect 2(4):225-33.
5. Taylor P, Gohel SR, **Di X**, Walter M, Biswal B (2012). Functional covariance networks: obtaining resting state networks from intersubject variability. Brain Connect 2(4):203-17.
4. Chan RC, **Di X**, McAlonan GM, Gong QY (2011). Brain Anatomical Abnormalities in High-Risk Individuals, First-Episode, and Chronic Schizophrenia: An Activation Likelihood Estimation Meta-analysis of Illness Progression. Schizophr Bull 37(1):177-88.
3. **Di X**, Chan RC, Gong QY (2009). White matter reduction in patients with schizophrenia as revealed by voxel-based morphometry: an activation likelihood estimation meta-analysis. Prog Neuropsychopharmacol Biol Psychiatry 33(8):1390-1394.
2. Chan RC, Huang J, **Di X** (2009). Dexterous movement complexity and cerebellar activation: a meta-analysis. Brain Res Rev 59(2):316-323.
1. Rao H, **Di X**, Chan RC, Ding Y, Ye B, Gao D (2008). A regulation role of the prefrontal cortex in the fist-edge-palm task: evidence from functional connectivity analysis. Neuroimage 41(4):1345-1351.

Publications in Chinese

3. Liu Y, Chen S, Fan F, **Di X**, Fan H, Feng C, Guo S, Gan Y, Li H, Lv X, Ren Z, Xu P, Yuan B, Zuo X, Hu C (2021): A standardized checklist on meta-analysis reporting in the open science era. Scientia Sinica Vitae doi:10.1360/SSV-2021-0009.
2. Hu C, **Di X**, Li J; Sui J, Peng K (2015). Meta-analysis of Neuroimaging Studies. Advances in Psychological Science 23(7): 1118-1129.
1. **Di X**, Rao H (2007). Progress in Functional Connectivity Analysis. Progress in Biochemistry and Biophysics 34(1), 5-12.

File Drawer (No plan for submission)

Hafiz R, Gandhi TK, Misrhra S, Prasad A, Mahajan V, Natelson B, **Di X**, Biswal BB (2022). Assessing functional connectivity differences and work-related fatigue in surviving COVID-negative patients. bioRxiv:10.1101/2022.02.01.478677

Di X, Biswal BB (2016): Sex-dependent and sex-independent brain resting-state functional connectivity in children with autism spectrum disorder. bioRxiv doi: http://dx.doi.org/10.1101/038026

TEACHING AND MENTORSHIP

Undergraduate Courses

- BME 210: Processing Fundamentals for Biological Signals
  2025 Fall, 2025 Spring, 2024 Fall
- BME 471: Principles of Medical Imaging
  2023 Fall

Directed student learning

- Doctoral Advisory Committee Member. Donna Chen: “Investigating the neurovascular correlates of cognitive function in individuals with spinal cord injury using functional near-infrared spectroscopy”, 2024.
- Doctoral Advisory Committee Member. Rakibul Hafiz: "Assessing structural and functional brain alterations and work-related fatigue in non-hyposmic and hyposmic COVID-19 survivors", 2022.
- Master’s Thesis Committee Member. Berk Can Yilmaz: “Comparison of longitudinal changes in resting state functional magnetic resonance imaging between Alzheimer’s and healthy controls’, 2020.
- Doctoral Advisory Committee Member. Azeezat Azeez: "Developmental and sex modulated neurological alterations in autism spectrum disorder", 2019.
- Doctoral Advisory Committee Member. Keerthana Deepti Karunakaran: "A multimodal approach to investigate brain reorganization after spinal cord injury using functional magnetic resonance imaging and functional near-infrared spectroscopy", 2019.
- Master's Thesis Committee Member. Rakibul Hafiz: "Subject and group level changes and comparison in functional connectivity under low vs. high cognitively demanding naturalistic viewing conditions using fMRI", 2017.
- Master's Thesis Committee Member. Dhruti Patel: "Effect of scan time on resting state parameters", 2013.
- Master's Thesis Committee Member. Hossein Ebrahimi Nezhad: "Differentiating schizophrenic patients from healthy control: application of machine learning to resting state fMRI", 2013.

INVITED TALKS & WORKSHOPS

Educational courses

OHBM educational course: Molecular Imaging: a Hands-on Tutorial based on Open Access Datasets FDG-PET: Data Processing and Analysis Using SPM	Seoul, Korea June 2024

OHBM educational course: Making Quality Control Part of Your Analysis: Learning with the FMRI Open QC Project QC for resting-state and task fMRI in SPM	Montreal, Canada July 2023
Preconference workshop for 6th biennial conference on resting-state and brain connectivity Single Subject and Group Analysis	Montreal, Canada September 2018
Preconference workshop for 4th biennial conference on resting-state and brain connectivity Physiophysiological interaction (PPI), Granger causality analysis, and dynamic causal modeling (DCM) for resting-state fMRI	Boston, MA, USA September 2014
OHBM educational course: resting-state brain networks Case Study: Single Subject and Group Analysis	Seattle, WA, USA June 2013
OHBM educational course: resting-state brain networks Case Study: Single Subject and Group Analysis	Beijing, China June 2012

Invited Talks

Molecular Connectivity Online Series Comparing Intra- and Inter-individual Correlational Brain Connectivity from Functional and Structural Neuroimaging Data	Online March 2025
Psychoradiology Online Series Time-varying and stationary brain connectivity during movie watching	Online January 2025
Symposium: What is brain connectivity? Inter-individual estimations of functional connectivity	Munich, Germany May 2024
Resting-State Brain Connectivity 2023 Time-varying and stationary brain connectivity during movie watching	Dallas, TX, USA September 2023
PET for brain connectivity: back to the future? Workshop Glucose metabolic covariance connectivity	Glasgow, UK May 2022
Neurochat 2020 Online Conference Estimations of the weather effects on brain functions using functional MRI - a cautionary tale	Online April 2020
Department of Management, Jinan University Brain functional connectivity during task-stats as revealed by fMRI	Guangzhou, China January 2020
BrainHack Global NYC 2018 Measuring task modulated connectivity from fMRI data using psychophysiological interaction and beta series	NYC, USA May, 2018
Department of Biomedical Engineering, Shenzhen University	Shenzhen, China

Task modulated brain connectivity using fMRI: method considerations and new findings	January 2018
Department of Biomedical Engineering, University of Electronic Science and Technology of China Task modulated brain connectivity using fMRI: method considerations and new findings	Chengdu, China January 2018
4th biennial conference on resting-state and brain connectivity Modulatory interactions of resting-state functional connectivity	Boston, MA, USA September 2014
NJIT Biomedical Engineering Seminars Modulatory interactions of resting-state brain functional connectivity	Newark, USA September 2014
NJIT Biomedical Engineering Seminars Task vs. Rest - Different Network Configurations between the Coactivation and the Resting-State Brain Networks	Newark, USA April 2013
Institute of Psychology, Chinese Academy of Sciences Neuropsychology and Applied Cognitive Neuroscience Lab The infrastructure of brain functions - from structure to physiology	Beijing, China June 2012
Center for Brain Health, UT Dallas NeuroPsychometric Research Lab The infrastructure of brain functions - from structure to physiology	Dallas, TX, USA April 2012

Conference Presentations

21. Yang H, **Di X**, Biswal BB. Aberrant whole-brain task-modulated connectivity in Schizophrenia under stop signal task. Poster presentation at the Sixth Biennial Conference on Resting State / Brain Connectivity, Montreal, Canada (2018).
20. **Di X**, Biswal BB. Assessing task related brain connectivity in a fast event-related designed stop signal task using psychophysiological interaction and beta series correlation. Poster presentation at the 4th New York Metropolitan Imaging Symposium, New York, USA (2017).
19. Azeez AK, **Di X**, Biswal BB. Biological Sex Modulations on Cortical Thickness in Autism Spectrum Disorder: An analysis of Autism Brain Imaging Data Exchange II. Poster presentation at the International Meeting for Autism Research (IMFAR), San Francisco, USA (2017).
18. **Di X**, Biswal BB. Task related brain networks derived from trial-by-trial variability of a slow event-related designed Flanker task. Poster presentation at the Fourth Biennial Conference on Resting State / Brain Connectivity, Boston, USA (2014).
17. **Di X**, Yuan R, Biswal BB. Modulatory interactions between the thalamus and visual cortex in resting-state are modulated by eye open/closed conditions. Poster presentation at the Fourth Biennial Conference on Resting State / Brain Connectivity, Boston, USA (2014).
16. Fu Z, **Di X**, Chan SC, Hung YS, Biswal BB, Zhang Z. Characterizing temporal variations of functional connectivity in resting-state. Poster presentation for Joint Annual Meeting ISMRM-ESMRMB, Milano, Italy (2014).

15. Fu Z, **Di X**, Chan SC, Hung YS, Biswal BB, Zhang Z (2013). Time-varying correlation coefficients estimation and its application to dynamic connectivity analysis of fMRI. Conf Proc IEEE Eng Med Biol Soc. 2013:2944-2947.
14. Zhang Z, Fu Z, Chan SC, Hung YS, Motta G, **Di X**, Biswal BB. Conference Paper: Adaptive window selection in estimating dynamic functional connectivity of resting-state fMRI. 9th International Conference on Information, Communications and Signal Processing (ICICS2013).
13. **Di X**, Biswal BB. Identifying the Default Mode Network Structure Using Dynamic Causal Modeling on Resting-state fMRI. Poster presentation at 19th Annual Meeting of the Organization for Human Brain Mapping, Seattle, USA (2013).
12. **Di X**, Biswal BB. The nonlinear intrinsic brain networks - modulations on resting-state functional connectivity by other regions. Poster presentation at 19th Annual Meeting of the Organization for Human Brain Mapping, Seattle, USA (2013).
11. **Di X**, Fu Z, Zhang Z, Chan SC, Biswal BB. Transient connectivity changes during a visual task - time-varying correlation estimation analysis. Poster presentation at 19th Annual Meeting of the Organization for Human Brain Mapping, Seattle, USA (2013).
10. Yuan R, **Di X**, Kim EH, Barik S, Rypma B, Biswal BB. Regional Homogeneity of Resting-state fMRI Contributes to Both Neurovascular and Task Activation Variations. Poster presentation at 19th Annual Meeting of the Organization for Human Brain Mapping, Seattle, USA (2013).
9. **Di X**, Kannurpatti SS, Rypma B, Biswal BB. Calibrating BOLD fMRI activations with neuro-vascular and anatomical constraints. Poster presentation at 18th Annual Meeting of the Organization for Human Brain Mapping, Beijing, China (2012).
8. Gohel S, **Di X**, Biswal BB. Trajectories of functional brain networks connectivity over life-span brain development. Poster presentation at 18th Annual Meeting of the Organization for Human Brain Mapping, Beijing, China (2012).
7. Taylor P, Gohel SR, **Di X**, Walter M, Biswal BB. Functional covariance networks: obtaining resting state networks from intersubject variability. Poster presentation at 18th Annual Meeting of the Organization for Human Brain Mapping, Beijing, China (2012).
6. Jin H, Wang P, **Di X**, Ye Z, Xu G, Mo L, Lin C, Rao H. Activation of Medial Prefrontal Cortex during Sport-related Anticipation: An fMRI Study. Poster presentation at 17th Annual Meeting of the Organization for Human Brain Mapping, Québec, Canada (2011).
5. Zhu S, **Di X**, Jin H, Wang P, Mo L, Zhou K, Zhuo Y, Rao H. Training shapes Cerebellum and parieto-frontal network in professional badminton players. Poster presentation at Annual Meeting of ISMRM, Montreal, Canada (2011).
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